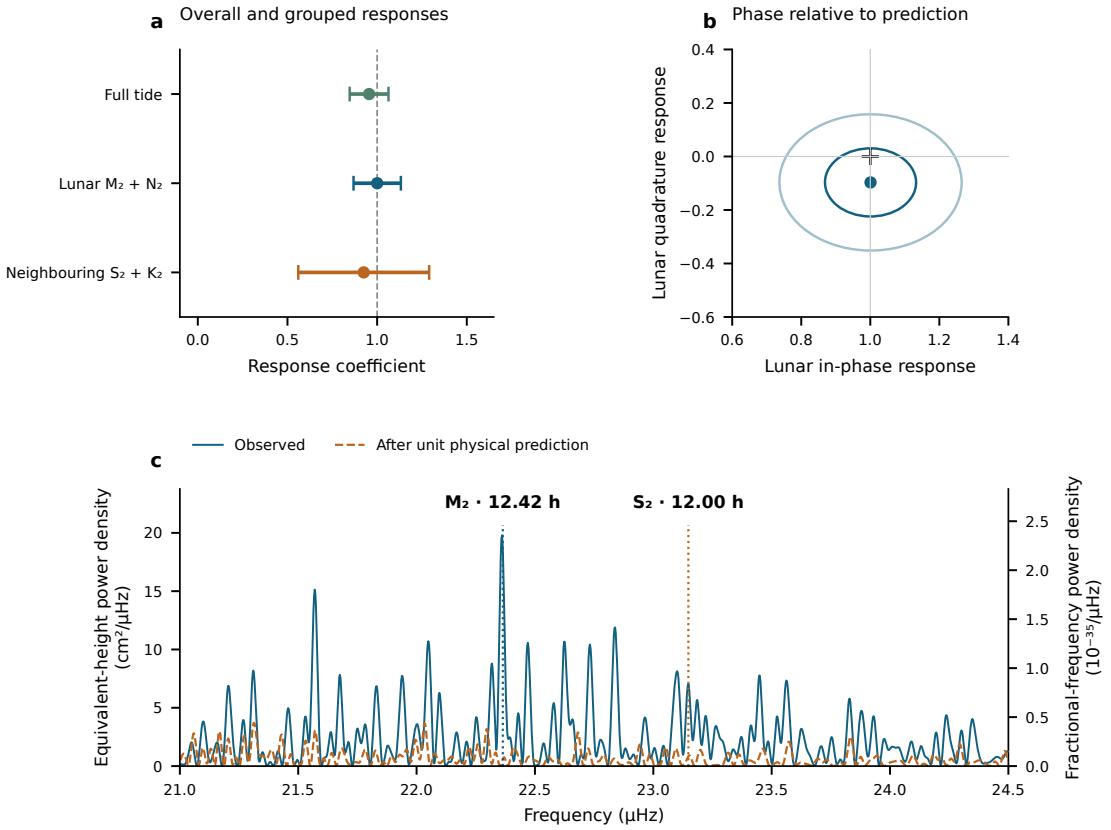
Tidal potential measured on a coastal–inland optical-clock baseline

Abstract — physical narrative, draft 2

The Moon and Sun change Earth's gravitational potential through their tidal fields and the response of the solid Earth and oceans. These changes alter the relative ticking rates of clocks at different locations. Here we detect tidal variations in the potential difference between coastal Shanghai and inland Wuhan using strontium and ytterbium clocks 692 km apart. The comparison has a reported fractional-frequency instability of $5.4 \times 10^{-16}/\sqrt{\tau/(1\,\mathrm{s})}$, where τ is averaging time. The predicted principal lunar signal has a fractional-frequency amplitude of 4.78×10^{-18}, equivalent to a height difference of about 4.4 cm. From 545.9 hours of observations spanning 353 days, we find an overall tidal response of 0.96 ± 0.11 relative to the combined solid-Earth and ocean-loading prediction (one statistical standard error). The measured spectrum shows distinct peaks at 22.361 and 23.147 μHz, near the principal lunar M_2 and solar S_2 frequencies, respectively. The lunar response also agrees with the predicted phase. These observations show that optical clocks can sense the changing gravitational environment over hundreds of kilometres and distinguish neighbouring tidal rhythms within it.

Figure 3 with equivalent-height units



Display-only revision. Panel **c** expresses the existing potential spectrum as equivalent-height power density. With conventional standard gravity $g_0 = 9.80665\,\mathrm{m/s^2}$,

$$S_{h,\mathrm{cm}} = 10^4 S_W / g_0^2 = 10^4 c^4 S_y / g_0^2.$$

The left axis is in cm²/μHz; the right axis retains fractional-frequency power density. Equivalent height represents a potential difference, not vertical station motion. The spectrum, peak locations, observing window and statistical results are unchanged. Panels **a** and **b** retain their existing dimensionless response coefficients and phase geometry.